

Motadarek: Audio-Radar Fall Detection and Severity Estimation System

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ABSTRACT Falls are a serious risk for elderly individuals and people living alone, where delayed assistance can lead to severe health complications. While many existing solutions focus on identifying whether a fall occurred, determining the level of danger remains difficult, especially when relying only on sound. Similar-sounding impacts may correspond to very different situations depending on the event's surrounding conditions. This paper presents Motadarek, a privacy-aware system for fall analysis that combines acoustic processing, contextual understanding, and radar-based post-fall monitoring. Instead of relying on a single direct severity predictor, the system uses multiple specialized components. These components identify the fall event, body posture, impact surface, distress-related vocalizations, and post-fall activity. Their outputs are combined through a transparent scoring mechanism to estimate whether the situation represents low or high risk. Experimental evaluation showed strong performance across the proposed components, including 0.99 macro F1-score for fall detection, 0.94 for position classification, 0.99 for surface classification, 0.94 macro F1-score for scream detection, and 0.91 macro F1-score for radar-based recovery analysis. These findings suggest that contextual and post-event cues can support severity estimation when impact audio alone is ambiguous. In addition, the modular structure allows the system to continue functioning even when some signals are unavailable. Overall, the proposed approach provides a practical and interpretable solution for non-intrusive emergency monitoring in real-world environments.

INDEX TERMS Acoustic monitoring, Audio Spectrogram Transformer (AST), Decision-level fusion, Contextual modeling, Radar sensing, Emergency monitoring systems.

I. INTRODUCTION

A. INTRODUCTION

Falls are defined as “events which result in a person coming to rest inadvertently on the ground or floor or other lower level” [1]. They are a leading cause of injury among individuals living alone, where delayed assistance following a fall can significantly increase the likelihood of injury and medical complications. This issue is especially pronounced among elderly individuals due to their increased vulnerability to falls and their long-term physical and psychological consequences. Consequently, timely fall detection and response are essential to improve safety and reduce potential health risks. In Saudi Arabia, falls remain a significant concern among the aging population. Studies indicate that approximately 21% of older adults in the Riyadh region experience at least one fall each year [2]. Despite this prevalence, many existing monitoring systems primarily focus on detecting whether a fall has occurred, without providing insight into its severity. As a result, caregivers often lack the critical information needed to assess urgency and prioritize response, which can limit the effectiveness of intervention.

Many existing fall monitoring approaches continue to face limitations in privacy, user compliance, and reliability, driving

increased interest in non-intrusive monitoring methods, such as audio-based sensing. While audio analysis offers a contact-free and less privacy-invasive approach to fall detection, relying solely on acoustic signals may not fully capture the context and severity of a fall.

This work proposes a modular multimodal system for detecting fall incidents and estimating their severity using audio-based analysis supported by contextual sensing. The proposed framework combines acoustic information with complementary cues derived from radar-based motion analysis and distress-related audio signals such as screams. These inputs are integrated via an interpretable, rule-based decision-fusion mechanism to estimate fall severity based on observable post-fall behavior rather than clinical assessment. The challenges and limitations motivating this work are discussed further in the following subsection.

B. PROBLEM DEFINITION

Individuals living alone, particularly elderly adults and people with mobility limitations, face serious risks when a fall occurs without immediate assistance. The primary concern is not only the fall itself, but the delay in detecting it, which can lead to severe complications or even life-threatening

outcomes. Although various monitoring technologies have been proposed, many are not practical in real-world settings. Wearable devices require consistent usage, while camera-based systems raise significant privacy concerns, especially in sensitive environments such as bedrooms and bathrooms. These concerns are particularly important in regions such as Saudi Arabia, where privacy within the home is highly valued. Non-intrusive sensing methods, such as audio-based and radar-based approaches, offer a promising alternative; however, existing solutions remain limited.

Most approaches focus only on detecting whether a fall has occurred, without assessing its severity. In addition, some advanced systems, such as those based on Doppler signals, rely on specific environmental conditions and are sensitive to acoustic reflections and environmental changes. They may also struggle in multi-person scenarios and often depend heavily on post-fall behavior for severity estimation, which may not always reflect the true level of risk. Therefore, there is a need for a reliable, privacy-preserving system that can not only detect fall events but also assess their severity based on the context of each fall, while maintaining robust performance in realistic home environments. This gap motivates the approach proposed in this work.

C. AIMS AND OBJECTIVES

The main objectives of this work are summarized as follows:

1. To design a privacy-preserving acoustic framework for fall detection, enabling reliable identification of fall events without relying on cameras or wearable devices.
2. To develop an interpretable severity estimation framework that models fall-related context, including body position and surface type, to support structured reasoning about fall events.
3. To propose a multi-evidence decision mechanism that integrates contextual information and post-fall cues (e.g., motion patterns and distress signals) through decision-level fusion.
4. To develop a precautionary severity estimation framework capable of providing early fall-risk assessment using immediate acoustic evidence, while remaining compatible with additional post-fall indicators when available.

To achieve these objectives, it is important to examine existing approaches in fall detection and severity estimation, particularly in audio-based and multimodal systems. The rest of the paper is organized as follows: In Section II, we review existing fall detection systems, audio-based approaches, radar-based post-fall monitoring, scream detection, and multimodal fusion techniques. In Section III, we describe the proposed pipeline, including the audio-based models, the scream detection model, the radar-based severity model, and the rule-based severity fusion strategy.

In Section IV, we present the experimental setup and model comparisons used to select the framework's final components. In Section V, we report the performance of the fall detection,

contextual audio classification, scream detection, and radar-based severity models. Finally, in Section VI, we present the main findings, limitations, and future directions of the proposed system.

II. LITERATURE REVIEW

A. FALL DETECTION SYSTEMS

A study presented in [3] categorized fall detection systems based on sensing technologies, including wearable sensors, camera-based systems, depth sensors, infrared devices, radar, and acoustic sensors. Despite advances in sensing technologies and machine learning, many systems still face challenges related to reliability, false alarms, user compliance, sensor placement, and performance in real-world environments. A wearable-based fall detection system has been proposed in [4], which combined a coarse-fine Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) to analyze inertial sensor data from the FallAID dataset, achieving an accuracy of 98%.

However, wearable systems rely heavily on user compliance and mainly focus on fall detection rather than severity assessment. Thus, a thermal-based fall detection system was presented in [5], which used low-resolution infrared sensors with Long Short-Term Memory (LSTM) and Gated Recurrent Unit GRU models, achieving about 0.93 sensitivity. Nevertheless, performance may be affected by environmental heat sources, increasing the risk of false alarms. A computer vision-based fall detection system was introduced in [6], which used OpenCV techniques, including posture monitoring, background subtraction, and contour analysis. However, performance may decline in complex or crowded environments.

On the other hand, an acoustic-based fall detection system was proposed in [7], which used inaudible acoustic signals and Doppler effect analysis to detect falls and estimate severity. Despite its promising results, the method requires specialized hardware and may be affected by environmental noise and multi-person scenarios. The radar-based monitoring system has also demonstrated potential for privacy-preserving fall detection. A Continuous Wave (CW) radar-based system discussed in [8] captured free-fall motion signatures and distinguished fall events from daily activities under indoor conditions. Despite the diversity of sensing modalities, many systems still struggle with reliability and real-world deployment challenges.

B. AUDIO-BASED FALL DETECTION

Recent advances in deep learning have shown strong capability in capturing temporal dependencies in sound data. One transformer-based approach [9] represents a significant development in this direction, as it learns global patterns from audio signals and achieves strong performance across various audio classification tasks. This makes it a suitable foundation for modeling complex acoustic events such as falls. Studies on audio-based fall detection have explored different approaches to identify fall events from environmental

sounds. One study [10] addressed this problem by analyzing environmental sounds and comparing different modeling approaches, showing that deep learning models outperform traditional methods in recognizing fall events. However, it treated the problem as simple fall detection and did not consider understanding the nature or severity of the fall.

Similarly, another study [11] investigated the problem using ambient sound and a transformer-based model, proposing a non-wearable, non-intrusive solution suitable for real-world environments such as bathrooms. The approach leverages temporal patterns in audio and performs well even with limited data. However, it focuses only on binary detection, does not model fall severity, and relies on a dataset collected in bathroom environments, which may limit generalization to other real-world settings.

C. RADAR-BASED POST-FALL MONITORING FOR SEVERITY ASSESSMENT

In recent years, researchers have begun to emphasize post-fall monitoring for its significant contribution to fall risk assessment. As proposed by Life Threatening-Fall (LT-Fall) [12], a two-stage mmWave radar approach is used to detect life-threatening falls. In the first stage, the system uses 4D point-cloud data to detect falls and fall-like events by monitoring sudden height drops. In the second stage, it verifies whether the event is life-threatening by checking if the person remains lying on the ground for a prolonged period. To support this verification, the system applies micro-motion localization to detect small body movements, such as breathing, when the person is motionless after the fall.

Similarly, researchers in [13] used a low-cost CW Doppler radar and a four-stage hierarchical algorithm to detect falls and track post-fall respiration rates. This approach shows that post-fall monitoring can also include physiological indicators, not only body posture or movement. By monitoring breathing after a detected fall, the system provides additional information about the person's condition after the event.

Furthermore, Ahn et al. [14] used 4D imaging radar point cloud data with a CNN to classify human postures into standing, sitting, and lying. Before training, the point-cloud data were preprocessed by padding with fewer-point frames, reducing dense frames using k-means clustering, and sorting the points by their x-, y-, and z-coordinates to ensure a consistent input structure. Their system also used a rule-based fall detection method based on changes in $z_{\max}$ and velocity to distinguish actual falls from normal lying-down movements. In addition, they introduced a Unity-based avatar visualization system for privacy-preserving real-time monitoring.

Overall, these studies demonstrated that different monitoring techniques, including 4D point-cloud analysis for assessing posture and inactivity, micro-motion localization for detecting breathing, and hierarchical algorithms for tracking respiration, can be integrated to estimate fall severity using both behavioral and physiological indicators. By combining measures such as inactivity duration, recovery time, breathing status, and sustained lying posture, these approaches provide

complementary methods for comprehensive post-fall assessment.

D. SCREAM RESEARCH IN DISTRESS DETECTION

In recent years, deep learning has been widely used for detecting human screams and distress signals, especially in real-world environments. A study in [16] proposed a hybrid CNN-LSTM architecture for identifying emergency-related sounds and triggering real-time alerts. Building on these hybrid approaches, another study in [17] integrated Wav2Vec2-based feature extraction with bidirectional LSTM networks to enhance scream detection and provide additional contextual cues, such as source localization, even in high-noise industrial settings. More recently, a CNN-Transformer framework was introduced in [18], allowing not only scream detection but also temporal interval estimation, thereby improving performance under noisy acoustic conditions.

In addition, a study in [19] explored the use of Unmanned Aerial Vehicle (UAV)-based acoustic systems for search-and-rescue scenarios. The proposed system combined signal processing techniques with deep learning models, such as Convolutional Recurrent Neural Networks (CRNNs), to detect distress signals in environments where visual sensing is unreliable. This is particularly useful in situations where victims are not visible due to obstacles or environmental conditions.

However, most existing approaches still treat scream or distress detection as a standalone task, in which the presence of a sound directly triggers an alert without considering its context. This limits the system's ability to make reliable decisions, particularly in real-world scenarios where multiple signals may be involved. In practice, distress-related sounds are often associated with other events, such as falls, yet current methods do not incorporate such contextual relationships or additional sensory evidence.

This highlights the need for more integrated approaches that move beyond isolated detection. In particular, combining scream detection with fall detection and severity estimation via decision-level fusion can provide a more reliable, context-aware understanding of emergency situations.

E. MULTIMODAL FALL DETECTION

Recent studies have explored multimodal fusion to improve reliability in fall detection by combining complementary information from multiple independent modalities. This reduces the uncertainty that occurs when each modality is used alone.

Several works emphasize decision-level fusion, where each modality is processed independently and the final outputs are combined at the decision stage. The study in [20] proposed a decision-level fusion framework using vibration and infrared modalities, applying Dempster-Shafer (D-S) evidence theory to resolve conflicts between sources. Similarly, the work in [21] highlighted that modality-specific architectures combined through decision-level fusion can improve predic-

tion reliability and support context-aware decision-making in healthcare monitoring.

In fall detection, single-source methods such as audio, scream detection, or radar capture only part of a fall event. This makes them vulnerable to background noise, dropped objects, or incomplete motion patterns. Multimodal fusion addresses this limitation by allowing independent classifiers to validate one another before a final decision is made.

A direct example of decision-level fusion in fall detection was presented in [22]. The authors combined a video stream using You Only Look Once version 7 Pose (YOLOv7-Pose) with a two-stream Spatio-Temporal Graph Convolutional Network (ST-GCN), and an audio stream using Mel spectrograms processed through MobileNetV2. The final outputs were fused using linear weighted summation and D-S evidence theory, with D-S fusion improving sensitivity compared to the video-only model.

Another decision-level approach was proposed in [23], where independent classifiers were trained on separate visual and audio feature streams, including Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), optical flow, ResNet-50 features, and Mel-Frequency Cepstral Coefficients (MFCC). Their outputs were combined using a Multilayer Perceptron (MLP), showing that audio can support fall detection when visual performance decreases under poor lighting.

Other studies combine decision-level fusion with feature-level fusion. The work in [24] used feature-level concatenation and decision-level weighted averaging with Inertial Measurement Unit (IMU), electroencephalography (EEG), and infrared sensors from the Universidad Panamericana Fall Detection Dataset (UP-Fall). The authors showed that combining modalities outperformed unimodal baselines.

The authors in [25] used fusion between millimeter-wave (mmWave) radar and floor vibration to connect motion evidence with impact evidence and reduce false alarms from object drops. The study in [26] fused skeleton, motion, and accelerometer features using a Graph Spatio-Temporal Channel Attention Network (GSTCAN) stream and a bidirectional LSTM stream, reporting strong performance on the UP-Fall and University of Rzeszow Fall Detection Dataset (UR-Fall).

Similarly, the work in [27] fused radar, WiFi Channel State Information (CSI), and acoustic features using an attention-based fusion strategy. The authors in [28] used decision-level weighted fusion by assigning weights to each detection method based on its F1-score and combining their outputs through weighted summation.

Overall, the literature shows that decision-level fusion is useful when independent modalities must confirm one another before making a final decision. This work follows the same principle, but applies it to fall severity estimation rather than binary fall detection. Instead of fusing raw signals or deep features, the proposed system applies a hierarchical rule-based severity fusion pipeline in which four independent classifiers, namely surface type, body position, scream detection, and radar-based recovery, are combined through

a deterministic scoring mechanism. A high-severity alert is triggered only when multiple evidence sources collectively support it, ensuring that no single uncertain signal dominates the final decision. This design choice provides the basis for the methodology described in the following section.

III. METHODOLOGY

This section presents the proposed multimodal fall severity estimation framework, including the overall pipeline, audio-based models, scream detection model, radar-based model, and rule-based severity fusion strategy.

A. PIPELINE

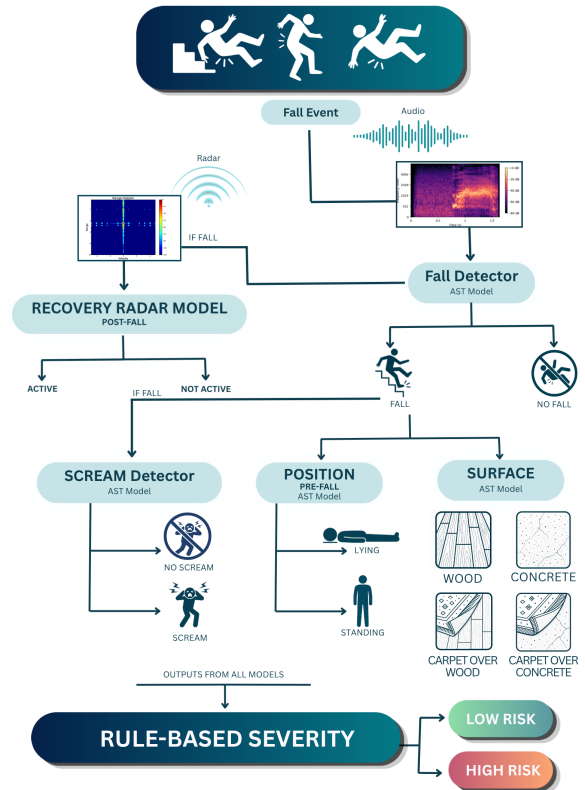


FIGURE 1. The proposed pipeline detects falls from audio and uses audio and radar cues to classify severity as low or high risk.

Figure 1 illustrates the overall architecture of the proposed multimodal fall severity estimation system. The pipeline is initiated when a fall event is detected. The acoustic signal is first passed to the Fall Detector AST, which serves as the primary gate of the entire system. If no fall is confirmed, the pipeline terminates immediately, and no further processing occurs. Only upon fall confirmation does the system proceed, activating the remaining models to build a contextual representation of the event.

Once a fall is confirmed, three audio models run in parallel on the same acoustic signal. The Position Classifier AST determines the pre-fall body posture, classifying it as Standing

or Lying. Simultaneously, the Surface Classifier AST identifies the impact surface as Concrete, Carpet-over-Concrete, Wood, or Carpet-over-Wood. The Scream Detector AST determines whether distress-related vocalization is present or absent. Concurrently, the Recovery Radar Model, Random Forest, analyzes post-fall motion, classifying the person's state as Active (recovered) or Not Active (remained still). All four models are engaged only following fall confirmation, ensuring that contextual assessment is conditional on a verified fall event. The outputs of the models then converge into the Rule-Based Severity System. Based on the estimated severity level, high-risk events trigger an automated caregiver alert. The following subsections present the models' architectures.

B. AUDIO-BASED MODELS

1) Pretrained AST Backbone

For all audio-based tasks, the Audio Spectrogram Transformer (AST) is adopted as the backbone model. AST is a deep learning architecture specifically designed for audio classification that operates on spectrogram representations of audio signals.

In this work, raw audio signals are first converted to log-Mel spectrograms, which serve as the model's input representation. Unlike traditional approaches that rely on Convolutional Neural Networks (CNNs), AST is a convolution-free architecture based entirely on Transformer layers and self-attention mechanisms. AST has demonstrated state-of-the-art (SOTA) performance across multiple audio classification benchmarks, outperforming several previous CNN-based and hybrid architectures.

2) Fall-Related Audio Models

- 1) **Data Collection** The fall-related models are trained using the SAFE (Sound Analysis for Fall Events) dataset [29], a publicly available audio dataset specifically designed for fall detection tasks. It contains 950 Waveform Audio File Format (WAV) recordings, sampled at 48 kHz and evenly balanced between fall and non-fall events. Each recording is approximately three seconds long and captures the full acoustic event.

Falls are simulated using a grappling dummy across various indoor environments that represent typical household conditions. The dataset includes different surface types such as concrete, wood, and carpet, along with realistic non-fall sounds such as walking and cleaning. The dataset is split into 10 folds to ensure consistent training and evaluation. File names follow a structured format:

AA-BBB-CC-DDD-FF.wav

Where each component is explained in Table 1.

The environment ID encodes important contextual attributes, particularly the surface type and the body position before impact. These factors significantly affect the severity of the fall and its impact on the person.

Falls on hard surfaces tend to be more severe, while

TABLE 1. SAFE Dataset File Naming Convention.

Code	Meaning
AA	Fold number (01–10)
BBB	Random numeric identifier
CC	Environment ID (surface, position)
DDD	Sequence number
FF	Class label (01 = fall, 02 = non-fall)

body position influences the dynamics and potential injury level. Therefore, these features are essential for accurate severity estimation in the proposed system.

2) Models

- a) **Fall Detection Model** For the fall detection task, a two-stage training strategy was adopted using the pretrained Audio Spectrogram Transformer (AST). In the first stage, Model A was trained only on the SAFE dataset to learn general acoustic representations for fall and no-fall events through binary classification.

To improve robustness under more diverse acoustic conditions, a second model (Model B) was initialized using the learned weights from Model A and then fine-tuned using both SAFE training samples and additional external fall-related audio recordings. These external samples included object-fall and impact sounds to help distinguish human falls from non-human impact events, reduce sensitivity to dataset-specific acoustic patterns, and improve handling of previously unseen fall sounds.

The raw audio signals were processed using the AST feature extractor, which converts waveforms into log-Mel spectrogram representations suitable for transformer-based audio analysis. The extracted representations were then passed through the AST backbone followed by a task-specific classification head for binary fall detection.

Training Setup:

Model A was trained using only SAFE training folds, while Model B was trained on a combined dataset containing SAFE recordings and external fall-related audio samples. The validation set for Model A consisted of SAFE validation folds, whereas Model B was evaluated on a mixed validation subset combining SAFE and external audio samples to better reflect the final training distribution.

Training was performed using the AdamW optimizer with a batch size of 8. Model A was trained for up to 10 epochs using a learning rate of 5×10^{-5} , while Model B was fine-tuned for up to 8 epochs using a smaller learning rate of 5×10^{-6} to preserve previously learned acoustic representations during adaptation. Early stopping with a patience value of 3 was applied to reduce

overfitting.

Model B was selected as the final fall detection component in the proposed pipeline due to its improved robustness under more varied acoustic conditions.

- b) *Position Classification Model* for the position classification task, the pretrained AST model is fully fine-tuned on the SAFE dataset using position labels (lying vs. standing). The raw audio signals are first processed using the AST feature extractor, which converts them into log-Mel spectrogram representations suitable for the model. These processed features are then used as input to the AST model.

Training Setup:

The model was trained for 15 epochs with a batch size of 8 using the AdamW optimizer with a learning rate of 3×10^{-5} . Cross-entropy loss was used as the objective function for classification. The AST backbone was fully fine-tuned for the position classification task. Training was conducted on a Graphics Processing Unit (GPU) when available.

- c) *Surface Classification Model* for the surface classification task, the pretrained AST model is fully fine-tuned on the SAFE dataset using four surface labels: concrete, carpet over concrete, wood, and carpet over wood. The raw audio signals are first processed using the AST feature extractor, which converts them into log-Mel spectrogram representations suitable for the model. These processed features are then used as input to the AST model for surface classification.

Training Setup:

The model was trained for 15 epochs with a batch size of 8 using the AdamW optimizer with separate learning rates for the transformer backbone and classification head. A learning rate of 3×10^{-5} was used for the AST backbone, while 5×10^{-5} was used for the classifier. Weight decay was set to 1×10^{-4} .

To address class imbalance, class weights computed from the training distribution were applied to the cross-entropy loss function. The AST backbone was fully fine-tuned for the surface classification task.

C. DISTRESS DETECTION FROM SCREAM SOUNDS

1) Dataset Preparation

The scream detection model draws on two publicly available audio datasets. Dataset 1 is the Human Screaming Detection Dataset [30], which contains 1,583 scream and 1,545 non-scream recordings. Dataset 2 is the Audio Dataset of Scream and Non Scream [31], which is more imbalanced, with only 862 scream samples against 2,631 non-scream recordings. Both were captured at 44,100 Hz, though speaker identi-

ties and recording environments differ considerably across sources. Non-scream content includes ambient noise, background speech, and music.

Class imbalance and source size differences required a balancing step before training. For each source, the larger class was trimmed to match the smaller, yielding 1,545 samples per class from Dataset 1 and 862 from Dataset 2. Sources were then equalized against each other by drawing equal per-class counts from both. Let n_k^{bal} denote the balanced per-class count for source k :

$$n_k^{\text{bal}} = \min(\text{scream}_k, \text{non-scream}_k)$$

The final per-class count drawn from each source is:

$$n = \min(n_1^{\text{bal}}, n_2^{\text{bal}}) = \min(1545, 862) = 862$$

Thus, n scream and n non-scream samples are drawn from each source, producing $4n = 4 \times 862 = 3,448$ samples.

The final dataset is equally distributed across the four source-label strata, defined as $\{\text{dataset1}, \text{dataset2}\} \times \{0, 1\}$, with each stratum contributing 862 samples. Files that cannot be decoded are removed after an audio validity check. The final dataset is split into training, validation, and test sets at 80:10:10, with stratification by the source-label key to preserve all four strata.

2) Audio Preprocessing and Augmentation

All recordings are converted to mono by averaging channels and resampled from 44,100 Hz to 16,000 Hz using the SoX Resampler (SoXR) high-quality resampler, matching the AST pre-training sampling rate. Each recording is standardized to 4 seconds, corresponding to 64,000 samples.

Recordings shorter than this fixed length are zero-padded at the end. Recordings longer than the fixed length are cropped using an energy-based method, where Root Mean Square (RMS) energy is computed per frame, and the crop is centered on the highest-energy frame. The hop length is 256 samples, and the frame length is 512 samples. During training, random jitter is added around the highest-energy frame before computing the crop start. Validation and testing use deterministic cropping without jitter.

Because the final balanced dataset contains only 3,448 samples, augmentation is applied during training to reduce overfitting and prevent the transformer from memorizing specific recordings. The augmentations expose the model to slightly different versions of each clip across epochs, encouraging it to learn general scream-related acoustic patterns rather than dataset-specific details. Waveform and spectrogram-level augmentations are summarized in Table 2.

Waveform augmentations are applied stochastically and independently by the `augment_audio` function. Time stretching changes the waveform length and is followed by random cropping to restore length L . Frequency and time masking are applied after log-Mel feature extraction by the `spec_augment_ast` function, following the SpecAugment strategy.

TABLE 2. Training-only augmentation operations for the scream detection model.

Operation	Domain	Parameter Range	Prob.
Temporal shift	Waveform	Max shift $\pm 20\%$ of clip length	0.5
Gaussian noise	Waveform	Noise factor $\sim \mathcal{U}(0.002, 0.01)$	0.5
Amplitude scaling	Waveform	Scale $\sim \mathcal{U}(0.7, 1.3)$, clipped to $[-1, 1]$	0.5
Pitch shifting	Waveform	$n_{\text{steps}} \sim \mathcal{U}(-1.0, 1.0)$ semitones	0.3
Time stretching	Waveform	Rate $\sim \mathcal{U}(0.9, 1.1)$; cropped to L	0.3
Frequency masking	Log-Mel	Mask width $\sim \mathcal{U}(0, 16)$ frequency bins	0.5
Time masking	Log-Mel	Mask width $\sim \mathcal{U}(0, 64)$ time frames	0.5

3) Model Architecture

For scream detection, AST is used as the backbone model. The raw audio signals are first converted into log-Mel spectrogram representations using the AST feature extractor. These spectrograms are then passed to the AST model for binary classification. The original classification head of the pretrained AST model is replaced with a new two-class classification head for Non-Scream and Scream. The transformer backbone is initialized from the pretrained AudioSet checkpoint, allowing the model to benefit from general acoustic representations learned from large-scale audio data before being fine-tuned on the scream detection dataset.

Unlike the position and surface classification models, which use separate learning rates for the backbone and classification head because they are trained on the smaller SAFE dataset, the scream detection model uses one uniform learning rate for all parameters. This is because the scream dataset is larger and more diverse, allowing the full model to adapt to scream-related acoustic patterns while reducing the risk of overfitting to a very small dataset.

4) Training Setup

Training uses AdamW with a learning rate of 2×10^{-5} and weight decay of 1×10^{-4} . A cosine annealing scheduler is applied with an initial learning rate of 2×10^{-5} , a minimum learning rate of 1×10^{-6} , and $T_{\text{max}} = 15$.

The model is trained for up to 15 epochs with batch size 8 and cross-entropy loss. Gradient clipping with $\text{max_norm} = 1.0$ is applied after each backward pass. Early stopping monitors validation macro F1-score with patience 6, and the checkpoint with the highest validation macro F1-score is retained.

To maintain balanced source-label contribution during training, a WeightedRandomSampler is used. Each sample is assigned a weight inversely proportional to the number of training samples with the same source and class label. Sampling is performed with replacement.

After training, the decision threshold is selected on the validation set by sweeping $\tau \in \{0.30, 0.35, \dots, 0.70\}$. The selected threshold maximizes the scream-class macro F1-score while requiring scream-class precision to remain at least 0.75. The precision constraint prevents degenerate thresholds that maximize recall by predicting scream for nearly all inputs. The selected threshold τ^* is then applied to the held-out test set.

D. RADAR-BASED MODEL

1) Data Collection

For the radar-based severity model, recordings were selected from the Patient Activity Recognition with Radar sensors (PARrad) dataset [15]. PARrad is a publicly available radar dataset containing synchronized radar data and timestamped activity annotations for human movements and fall-related events in two different environments: a synthetic hospital room (Homelab) and a real-life hospital room (Hospital).

The dataset contains 24 subjects, each of whom performed 2-4 sets of recordings. Each recording set includes a raw radar file, `processed_data.h5`, and a related annotation file, `timestamp_speech.csv`, which contains the timestamps and commands of all events. The radar file includes data from two radar sensors: `TI_xWR14xx` and `TI_xWR68xx`. Each radar sensor provides three distinct representations: radar cube, micro-Doppler, and Range-Doppler map (RDN). For our work, we used only the RDN representation from the `TI_xWR68xx` radar because it provides a compact view of range and velocity information, which is suitable for analyzing post-fall motion.

A 15-subject subset of recordings was selected from PARrad for the radar severity experiment. After cleaning the data, the final radar severity dataset contained 117 valid across 12 subjects.

2) Proxy Severity Labeling

Since the PARrad dataset does not provide severity labels, proxy severity labels were derived using Post-Fall Recovery Time (PFRT). For each fall event, the timestamp of `fall on floor of room` was used as the fall time, and the first subsequent `stand up from floor of room` event was used as the recovery time. PFRT was the time difference between these two events.

Before processing the raw radar files, all available annotation Comma-Separated Values (CSV) files were analyzed to determine the distribution of PFRT across the dataset and identify a suitable threshold. This analysis was performed using only the timestamp annotations. The PFRT distribution showed that most falls were followed by rapid recovery, while a smaller proportion showed delayed recovery. Based on this distribution, a primary threshold of $T = 8$ seconds was selected to separate quick recovery from delayed recovery.

Labeling logic:

$$y_i(T) = \begin{cases} 0, & \text{if recovery is observed and PFRT}_i \leq T \\ 1, & \text{if recovery is observed and PFRT}_i > T \\ 1, & \text{if recovery is not observed and } H_i > T \\ -1, & \text{if recovery is not observed and } H_i \leq T \end{cases}$$

where $y_i(T)$ is the proxy severity label for fall event i . A label of 0 indicates a low-risk fall, 1 indicates a high-risk fall, and -1 indicates an excluded event. PFRT_i denotes the post-fall recovery time, and H_i denotes the remaining observation time from the fall until the end of the recording. Events labeled as -1 were excluded because the recording ended before enough post-fall time was available to determine whether recovery would occur after the threshold.

After cleaning, the final dataset for the main $T = 8$ experiment contained 117 valid fall events, including 87 low-risk and 30 high-risk samples.

3) Radar Window Extraction and Preprocessing

After proxy labels were assigned, a radar window was extracted around each fall event from the RDN sequence. The fall timestamp was converted into a radar frame index using the estimated frame rate of each recording:

$$fps = \frac{N}{t_{\text{end}} - t_{\text{start}}}$$

where N is the total number of RDN frames. For each fall, a window from 1 second before the fall to 15 seconds after the fall was extracted:

$$W_i = [t_i^{\text{fall}} - 1s, t_i^{\text{fall}} + 15s]$$

The pre-fall buffer captures the impact-related motion, while the post-fall segment captures recovery behavior. Each RDN frame is 93×128 in size, representing range and Doppler bins.

The extracted RDN windows were then preprocessed before feature extraction. The absolute value was taken to represent radar reflection strength, and log compression was applied to reduce the dominance of very strong reflections:

$$X = \log(1 + |W_i|)$$

To reduce the effect of static objects in the scene, the zero-Doppler region near the center Doppler bin was removed. This helps suppress reflections from non-moving objects such as walls, furniture, and the floor. After that, a temporal median background was computed from the RDN window:

$$B = \text{median}(X, \text{axis} = \text{time})$$

The background was then subtracted from the window, and negative values were clipped to zero:

$$X' = \max(X - B, 0)$$

This keeps the motion-related changes in the window while reducing static reflections. The resulting preprocessed RDN window was used for feature extraction.

4) Feature Extraction

From each preprocessed RDN window, 16 handcrafted features were extracted to describe post-fall motion. For each frame t , a number of intermediate quantities were computed from the preprocessed RDN frame X_t . $e(t)$ was first computed as the total residual energy after removing clutter and subtracting background. Then, the Doppler centroid $\mu_D(t)$ and Doppler spread $\sigma_D(t)$ were calculated from the Doppler profile to describe the velocity distribution in the frame. In addition, $r^*(t)$ was defined as the range bin with the maximum energy, and the spatial distribution of the energy over the range bins was described by the range spread $\sigma_R(t)$. The extracted features are shown in Tables 3, 4, 5.

TABLE 3. Motion-energy features extracted from each preprocessed RDN window.

Feature	Description
energy_mean	Mean motion energy across the window.
energy_std	Motion energy variation.
energy_min	Minimum motion energy.
energy_max	Maximum motion energy.
energy_slope	Motion energy trend over time.
stillness_ratio	Fraction of low-motion frames.
time_to_stillness	Time until first low-motion frame.

TABLE 4. Doppler-based features extracted from each preprocessed RDN window.

Feature	Description
doppler_centroid_mean	Mean Doppler centroid.
doppler_centroid_std	Doppler centroid variation.
doppler_spread_mean	Mean Doppler spread.
doppler_spread_std	Doppler spread variation.

TABLE 5. Range-based features extracted from each preprocessed RDN window.

Feature	Description
peak_range_mode	Most frequent peak range bin.
peak_range_std	Peak range variation across frames.
peak_range_changes	Peak range transition rate.
range_spread_mean	Mean range spread.
range_spread_std	Range spread variation.

5) Model Configuration and Training Setup

After feature extraction, all radar feature files were merged into a single dataset. Any event marked as excluded, skipped, or truncated was dropped prior to training. Applying a binary proxy-labeling threshold of $T=8$ seconds left 117 usable fall events in total: 87 classified as low-risk (label 0) and 30 as high-risk (label 1).

For classification, we built a Random Forest model drawing on all 16 extracted radar features. Random Forest was chosen for its ability to capture non-linear feature relationships,

which is particularly useful for distinguishing between quick and delayed recovery patterns.

To avoid data leakage between the training and test sets, a subject-based group split was used. All samples from a given subject remained in the same partition, making this a stricter evaluation than random sampling. The model is tested only on subjects it has never seen before. Training used 500 trees, a max depth of 4, a minimum leaf size of 2, and balanced class weights to handle the class imbalance. Features were standardized within the pipeline prior to training.

Because the dataset was imbalanced, a tuned decision threshold replaced the default 0.5 cutoff when converting output probabilities to class labels. Thresholds between 0.10 and 0.90 were tested, and 0.3 was selected for the final model. The lower threshold reflects the priority of catching delayed-recovery falls, as false negatives carry a higher cost than false alarms in this context.

E. RULE-BASED SEVERITY FUSION

TABLE 6. Severity Scoring Schema.

Factor	Context	Points	Rationale
Pre-fall position	Standing	+2	Greater fall height
Pre-fall Position	Lying	+1	Lower-height fall
Surface	Concrete / Carpet-over-Concrete	+3	Harder impact surface
Surface	Wood / Carpet-over-Wood	+1	Lower-risk surface
Scream	Scream detected	+3	Possible distress indicator
Scream	No scream detected	+0	Non-diagnostic
Radar	Not recovered after fall	+3	Possible inability to recover
Radar	Recovered after fall	+0	Lower immediate risk

All model outputs are integrated using a deterministic precautionary scoring mechanism that identifies potentially high-risk fall situations. The assigned scores do not represent medically validated injury probabilities; instead, they reflect the contextual risk importance inferred from prior fall analyses and biomechanical studies. Harder surfaces were assigned higher scores because previous studies reported more severe injuries on concrete-related flooring, while wooden subfloors resulted in fewer hip fractures than concrete subfloors [32]. Therefore, concrete and carpet-over-concrete surfaces were assigned higher risk scores than wooden-related surfaces.

Moreover, body position was also weighted differently because impact force increases with fall height. The study [33] reported that greater fall heights produce larger impact forces during forward falls. Since standing represents a higher fall height than lying, standing was assigned a higher score. The high-risk threshold was set to 5. Concrete-related surfaces were intentionally assigned a score of +3 so that a standing fall onto concrete or carpet-over-concrete surfaces can directly exceed the threshold and be classified as *High Risk* immediately, even without post-fall indicators. This allows rapid emergency prioritization during the fall event itself without waiting for additional confirmation signals.

Scream detection and radar inactivity after the fall were also assigned relatively high scores (+3) because they act as strong post-fall distress indicators. Their contribution becomes particularly important in lower-impact scenarios, such as falls on wooden surfaces, where distress signals or the inability to recover after the fall may still indicate a dangerous situation and elevate the final score beyond the threshold.

Overall, the threshold and weights were selected using a conservative precautionary strategy intended to prioritize potentially dangerous falls rather than provide a medical diagnosis. Table 6 summarizes the complete precautionary severity scoring strategy and the assigned contextual risk weights used in the proposed fusion framework. The next section presents the experimental setups and model-specific comparisons conducted to evaluate the proposed system components.

IV. MODELS EXPERIMENTS

A. PERFORMANCE METRICS

To ensure consistent evaluation across all classification tasks, model performance was assessed using Accuracy, Precision, Recall, and F1-score. These metrics provide a standardized evaluation of the proposed models across fall detection, scream detection, position classification, surface classification, and severity estimation tasks. Accuracy measures the proportion of correctly classified samples among all predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision measures the proportion of correctly predicted positive samples among all predicted positive samples:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall measures the proportion of actual positive samples that were correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1-score is the harmonic mean of Precision and Recall:

$$F1\text{-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

In addition, the macro F1-score was used to evaluate overall multi-class performance by averaging the F1-score equally across all classes.

B. SEVERITY ESTIMATION STRATEGY EVOLUTION

Initially, severity estimation was explored as a direct binary classification task using fall audio recordings labeled as High-Risk or Low-Risk through human listening and voting. The objective was to investigate whether fall severity could be inferred directly from impact sounds alone.

However, during experimentation, it became evident that relying solely on raw impact audio was insufficient for reliable severity estimation. Similar fall sounds could correspond

to different physical situations depending on contextual factors such as body position and surface type. These observations suggested that meaningful contextual information about the fall event could be better modeled using specialized audio classification tasks rather than a single end-to-end severity classifier.

Based on these findings, the proposed system transitioned to a modular, context-aware framework comprising specialized models for fall detection, position classification, and surface classification. Instead of directly predicting severity from raw audio, the system first extracts contextual information from the fall event itself.

In addition, reliable severity estimation also requires consideration of post-fall indicators, such as distress signals and post-fall activity, since fall severity cannot always be inferred from the impact event alone. Therefore, the proposed framework incorporates scream detection and radar-based activity monitoring alongside the audio-based contextual models. The outputs from these specialized components are combined through a rule-based decision-level fusion strategy to generate the final severity assessment. This approach provides a more interpretable, context-aware estimation process than direct audio-only severity prediction.

C. MULTI-HEAD VS SEPARATE TASK-SPECIFIC MODELS

To investigate whether shared contextual learning benefits the proposed framework, we compared a shared multi-head architecture against separate task-specific models for position and surface classification. In the multi-head approach, a single AST backbone was trained jointly for both tasks using separate output heads. In contrast, the task-specific approach used two independently trained AST-based models, where each model was dedicated to a single contextual classification task.

The task-specific models consistently outperformed the shared multi-head architecture across both contextual classification tasks, as shown in Tables 7 and 8. For position classification, the task-specific model achieved 95% accuracy, compared to 90% accuracy for the multi-head approach. Similarly, in surface classification, the task-specific model achieved 98% accuracy, outperforming the multi-head architecture, which achieved 93% accuracy.

These results suggest that independently trained models were more effective in learning specialized acoustic representations for each contextual task while reducing interference between heterogeneous classification objectives. This observation is consistent with the nature of the tasks, since body position and impact surface may depend on different acoustic characteristics. Training separate AST models allowed each model to specialize in task-specific acoustic patterns without being constrained by a shared representation space. Based on these observations, separate task-specific AST models were adopted in the final framework.

TABLE 7. Accuracy Comparison Between Shared Multi-Head and Task-Specific Architectures.

Architecture	Position Acc.	Surface Acc.
Multi-Head	0.90	0.93
Task-Specific	0.95	0.98

TABLE 8. Macro-Average Metric Comparison Between Shared Multi-Head and Task-Specific Architectures.

Architecture	Task	Macro Prec.	Macro Rec.	Macro F1
Multi-Head	Position	0.90	0.90	0.90
Multi-Head	Surface	0.95	0.93	0.93
Task-Specific	Position	0.97	0.91	0.94
Task-Specific	Surface	0.99	0.98	0.99

D. SCREAM DETECTION COMPARISON

The first set of experiments used classical machine learning models, where audio features were extracted manually rather than learned directly from data. Four classifiers were tested using an MFCC-based feature vector, with results reported in Table 9. Among the classical models, the Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel achieved the best performance, with an accuracy of up to 91%, while Logistic Regression, Random Forest, and Gradient Boosting produced similar accuracy scores with around 88-89%.

The CNN followed a different approach, learning directly from Mel-spectrogram representations rather than relying on handcrafted MFCC features. Although the CNN slightly outperformed the SVM, the improvement was limited, suggesting that training a model from scratch on a relatively small dataset constrained its ability to learn more general acoustic representations.

The AST model was therefore explored as a pretrained transformer-based alternative. Unlike the CNN, which starts from randomly initialized weights, AST benefits from prior acoustic knowledge learned during AudioSet pre-training. Given the limited training set size, this prior knowledge provided a clear advantage. As shown in Table 9, AST achieved the highest macro precision, macro recall, and macro F1-score of 0.94, and was selected as the final scream detection model.

TABLE 9. Performance Comparison of Scream Detection Models.

Model	Macro Prec.	Macro Rec.	Macro F1
Logistic Regression	0.88	0.88	0.88
RBF-SVM	0.91	0.90	0.90
Random Forest	0.89	0.88	0.88
Gradient Boosting	0.88	0.88	0.88
CNN	0.91	0.91	0.91
AST (Final)	0.94	0.94	0.94

E. RADAR-BASED MODEL COMPARISON

In this experiment, we compared multiple machine learning classifiers to choose the most suitable model for radar-based fall severity classification. All models we trained used the same radar-derived post-fall features and were evaluated using accuracy, macro precision, macro recall, and macro F1-score. As shown in Table 10, Random Forest achieved the best overall balance, with an accuracy of 92%, macro precision of 0.91, macro recall of 0.91, and macro F1-score of 0.91. Although Extra Trees achieved the same accuracy of 92% and a higher macro precision of 0.95, macro F1-score were lower at 0.90. Therefore, we selected Random Forest as our final classifier because it achieved higher macro recall and macro F1-score across the Low and High severity classes.

TABLE 10. Performance Comparison of Radar-Based Severity Classification Models.

Model	Acc.	Macro Prec.	Macro Rec.	Macro F1
Random Forest	0.92	0.91	0.91	0.91
Extra Trees	0.92	0.95	0.88	0.90
SVM-RBF	0.88	0.86	0.88	0.87
KNN	0.89	0.93	0.81	0.85
Gradient Boosting	0.89	0.93	0.81	0.85
Gaussian NB	0.84	0.82	0.82	0.82
AdaBoost	0.81	0.78	0.76	0.77
Logistic Regression	0.81	0.89	0.69	0.71

The following section presents the obtained results, including classification performance and confusion matrix analysis across the proposed system components.

V. RESULTS AND FINDINGS

A. FALL DETECTION MODELS PERFORMANCE

The fall detection module used a staged AST adaptation process, in which the final detector (Model B) was fine-tuned using both SAFE recordings and additional external fall audio samples to improve robustness across varied acoustic conditions. Model A achieved 99% validation accuracy on SAFE validation folds, while Model B obtained 99% accuracy on the mixed validation subset. The final evaluation of Model B achieved an F1 Score of 0.99 in both categories. Additional testing on external audio samples further showed that the detector could still recognize previously unseen fall sounds and differentiate between human and object falls despite differences in recording environments and the limited number of external samples.

The confusion matrix in Figure 2 shows that the final fall detection model maintained highly consistent predictions across both classes on the combined validation set. Most recordings from both classes were correctly identified, with only a small number of samples misclassified.

B. POSITION AND SURFACE CLASSIFICATION PERFORMANCE

The confusion matrix in Figure 3 shows that the position classification model achieved strong overall performance for both lying and standing classes. Most standing samples were

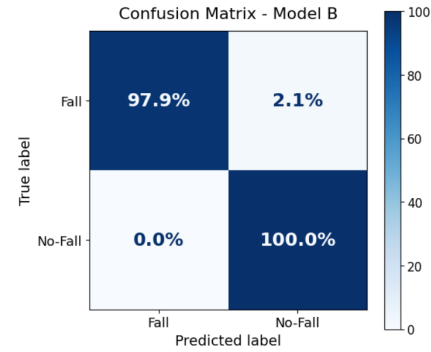


FIGURE 2. Confusion matrix of Model B on the mixed validation dataset.

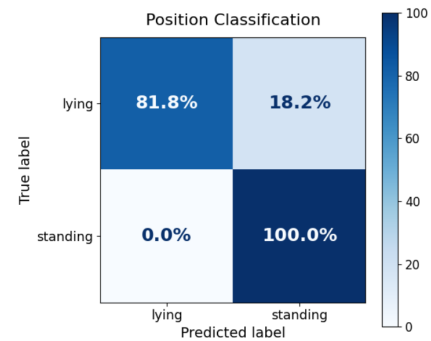


FIGURE 3. Confusion matrix for the position classification model.

correctly identified, while a small number of lying samples were misclassified as standing due to the smaller number of lying samples in the evaluation set. Nevertheless, the model maintained high recall for standing-related falls, which is particularly important because standing falls are generally associated with a higher risk of injury.

The confusion matrix in Figure 4 shows that the surface classification model successfully learned distinctive acoustic characteristics for different surface types, achieving highly accurate predictions across all classes. Only one concrete sample was misclassified as carpet over concrete, which is

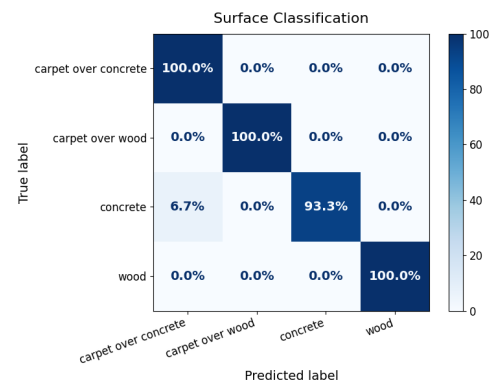


FIGURE 4. Confusion matrix for the surface classification model.

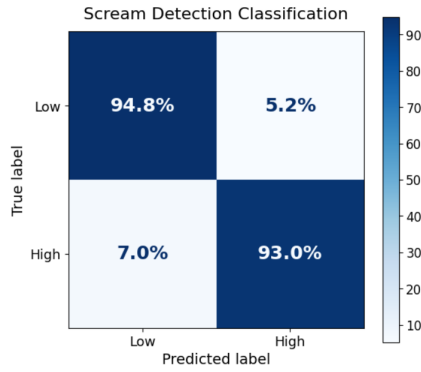


FIGURE 5. Confusion matrix for the scream detection model.

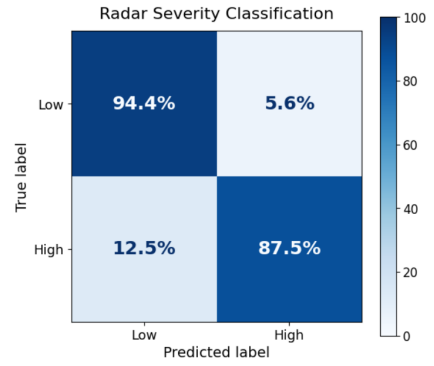


FIGURE 6. Confusion matrix for the radar-based severity classification model.

reasonable due to the shared concrete impact characteristics between the two surface conditions.

C. SCREAM DETECTION PERFORMANCE

Testing was carried out on 345 held-out samples using the selected threshold $\tau^* = 0.35$. Figure 5 shows the confusion matrix. Out of 173 Non-Scream samples, 164 were correctly identified, while 9 were incorrectly classified as Scream. For the Scream class, 160 out of 172 samples were correctly detected, leaving 12 missed screams. In an emergency monitoring system, missed screams are the more critical error because a person in distress going undetected is a worse outcome than generating unnecessary alerts. The threshold was therefore set to 0.35 to accept slightly more false positives while keeping missed screams low.

The model achieved an accuracy of 94% and a macro F1-score of 0.94. The ROC-AUC score of 0.98 indicates strong separation between the two classes, not only at the selected threshold but across different threshold values. This suggests that the model maintains reliable class discrimination under changing deployment conditions.

Per-source results show an expected performance gap between the two datasets. Dataset 1 achieved a macro F1-score of 0.99, with perfect scream precision and 0.99 recall. Dataset 2 was more challenging, achieving a macro F1-score of 0.88. This difference is expected because Dataset 2 has greater acoustic variability and was originally more imbalanced. Even on the more difficult source, however, performance remained above the ML and CNN baselines in Table 9, confirming the advantage of AudioSet pre-training for limited, diverse audio data.

D. RADAR SEVERITY MODEL PERFORMANCE

After comparing the classifiers' evaluation results, Random Forest was selected as the final radar-based severity classifier. The best configuration used a decision threshold of 0.30. The test set contained 26 fall samples, including 18 Low-severity falls and 8 High-severity falls. The model achieved an overall accuracy of 92% and a balanced accuracy of 91%, showing that it performed well across both severity classes.

As shown in the confusion matrix Figure 6, the model correctly classified 17 out of 18 low-severity falls; only one was misclassified, and 7 out of 8 high-severity falls were also correctly classified; only one was misclassified. This result indicates that the extracted radar-based post-fall features were useful for distinguishing between faster recovery cases and delayed recovery cases. Since high-severity falls are the most safety-critical class, achieving a high-severity recall of 0.88 is important because it means that most high-risk falls were correctly identified.

Overall, the obtained results support the feasibility of the proposed modular framework for fall monitoring beyond binary detection. The following section further discusses the contribution of the different audio and radar modules to contextual and post-fall severity interpretation.

VI. DISCUSSION

The results show that we can use the proposed modular framework to provide more informative fall monitoring than fall detection alone. Our audio-based fall detector achieved strong performance, with the final AST-based model obtaining an F1-score of 0.99 for both fall and non-fall classes. This suggests that transfer learning can effectively detect impact-related fall sounds, especially when fall audio data is limited. However, we should not overstate this result, as we evaluated the model on curated recordings rather than continuous real-world environments.

The position and surface classification models also showed that we can use audio to extract useful contextual cues about a fall event, such as body position and impact surface. These cues are important because fall severity may depend not only on whether a fall occurred, but also on the surrounding context. Similarly, the scream detection model added a distress-related cue, achieving 94% accuracy, a macro F1-score of 0.94, and a ROC-AUC of 0.98. We selected the threshold to prioritize reducing missed screams, which is suitable for safety-critical monitoring.

The radar-based severity model achieved 92% accuracy and 91% balanced accuracy using post-fall features. It correctly classified 17 out of 18 low-severity falls and 7 out of

8 high-severity falls. This indicates that we can use radar-based recovery behavior to distinguish between faster recovery cases and delayed recovery cases. Therefore, the radar module supports our goal of estimating fall risk level rather than only detecting the fall event.

The final risk level was determined by a decision-level severity scoring schema, giving an interpretable cue for each module. Audio provides pre-fall position, surface, and scream, radar post-fall recovery status. Cues associated with high risk (e.g., hard surfaces, detected screams, or failure to recover) are scored higher. A score of 5 or greater was considered high risk in cases. In the case of a high-risk classification, the system can alert the caregiver to enable a quicker response. This scoring design makes the final risk estimate more interpretable by showing which cues contributed to the high-risk decision.

Despite these promising results, our project has some limitations. The severity labels were based on environmental cues and recovery behavior rather than clinical injury outcomes. Therefore, we interpret the model as estimating risk level rather than diagnosing injury severity. Most importantly, we did not have a synchronized audio-radar dataset for the same fall events; therefore, we could not test the full pipeline end-to-end under the same fall scenarios.

Overall, the findings suggest that combining audio and radar cues is a promising direction for privacy-preserving fall monitoring. In this project, the fusion was modular and decision-level, since the audio and radar modules were trained separately rather than on fused audio-radar samples. Audio can support fast fall detection and distress recognition, while radar can provide post-fall motion cues to estimate severity. Future work should evaluate the full system using a synchronized audio-radar dataset and develop a true multimodal fusion model trained on paired audio-radar fall samples. Future studies could also use 4D mmWave radar point-cloud data to estimate changes in body height more directly, and extend the radar module with deep learning or transformer-based models to detect recovery actions such as standing up or remaining inactive. Finally, the system could be expanded to recognize high-risk fall-related conditions, such as epileptic seizures, fainting, or heart attack-related collapses, to provide more specific and clinically meaningful alerts.

VII. CONCLUSION

This work presents a modular multimodal framework for detecting falls and estimating their severity. The audio modules detect falls and provide context on body position, impact surface, and distress sounds. The radar module then checks post-fall movement to see whether the person recovered or stayed inactive.

The results show that the proposed modules can provide useful evidence for understanding fall events beyond simple binary detection. The AST-based fall detector achieved strong performance, while the position, surface, and scream detection models added important contextual information. The radar-based severity model also showed that post-fall

motion patterns can help distinguish between faster recovery and delayed recovery cases.

Experimental evaluation showed consistently strong performance across the proposed modules, with fall detection reaching 99% accuracy, position classification achieving 95%, surface classification reaching 98%, and the radar-based recovery analysis achieving 92% accuracy. Within the proposed framework, each model contributed a specific contextual cue for severity estimation: the position model identified the body position before the fall (standing or lying), the surface model estimated the impact surface type, the scream detector provided distress-related evidence, and the radar module analyzed post-fall recovery activity. These outputs were combined through a rule-based decision mechanism using precautionary severity rules derived from observable event conditions. This modular design allowed the severity estimation process to remain interpretable and easier to interpret and analyze compared to relying on a single direct prediction model.

Instead of treating severity as a direct medical diagnosis, we designed the system to estimate fall risk using observable evidence. This makes the framework more interpretable and suitable for privacy-preserving monitoring. However, the system still requires further validation using synchronized audio-radar data collected from the same fall scenarios. Future work should focus on end-to-end multimodal evaluation, stronger radar representations such as 4D mmWave point clouds, and more advanced models for detecting recovery actions and high-risk fall-related conditions.

APPENDIX A HUGGING FACE DEPLOYMENT

The proposed Motadarek system was deployed as an interactive Hugging Face Space to demonstrate the repository and fall severity estimation workflow. The interface allows users to access the implemented system through the following public link:

<https://huggingface.co/spaces/SarahSulaiman/Motadarek>

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